ELECTRICITY DEMAND FORECASTING INTELLIGENCE SYSTEM

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Problem and Research Gap

The decision problem is bigger than a single demand prediction

01 **Point forecasts hide operating risk**
A single value does not show the plausible best- and worst-case demand range.

02 **Static models can age**
Load relationships shift with weather, calendars, events and changing demand patterns.

03 **Numbers alone lack behavioural context**
Operators still need to know whether a day is normal, peak, low or genuinely abnormal.

04 **Separate tools weaken decision flow**
Forecasting, anomaly review, explanation, and planning should connect through one platform.

Four objectives turn load history into operational intelligence

01

Adaptive Forecasting

What will demand be—and how uncertain is it?

Forecast + interval

02

Behaviour Intelligence

What type of demand day is this—and is it abnormal?

Daily label + alert

03

Explainable AI

Which features drove the prediction?

Local + global explanation

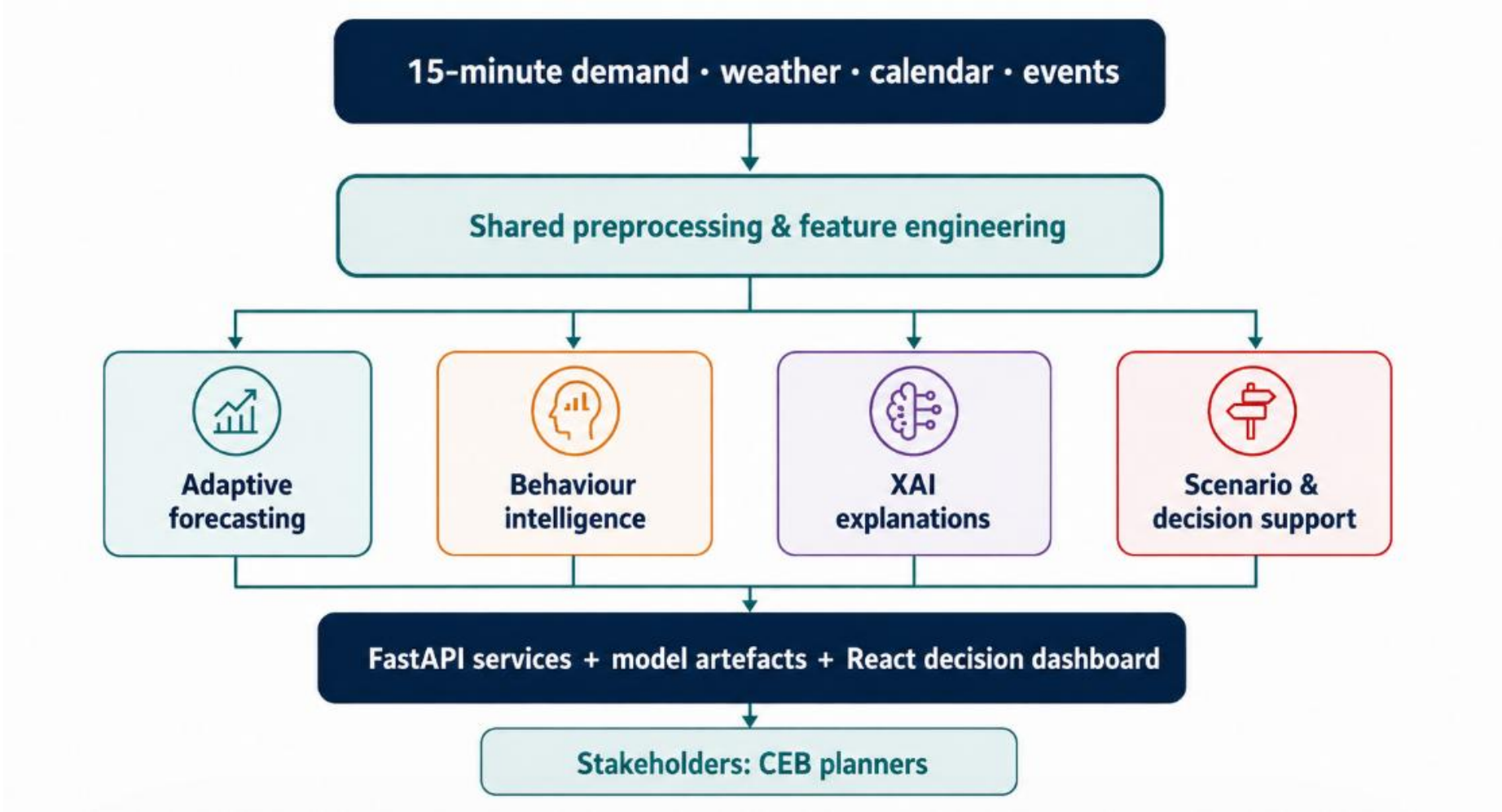
04

Decision Support

What happens under a changed condition?

Scenario comparison + action

System Overview



Integration & Completion Status

Component	Core Implementation	API Integration	Frontend Integration	Evaluation	Remaining Work
Adaptive Forecasting	100%	85%	77%	100%	Wire /api/predict and /api/compare into its own UI
Behaviour Intelligence	100%	100%	55%	100%	Add UI triggers for the 5 pipeline re-run endpoints
XAI	100%	100%	100%	100%	Add a /health endpoint for uptime monitoring
Decision Support	100%	100%	67%	100%	Reject empty request bodies instead of returning 0.0 kW

Non-Functional Requirements

Forecast API response time

~4 ms

Behaviour analysis response time

~86 ms

MC Dropout inference (100 passes)

~0.26 s

Dashboard initial load time

~0.6 s

Concurrent API requests supported

10 / 10

Non-Functional Requirements

Passed tests

18 / 18

Code coverage

Not measured

Empirical 95% interval coverage

90.8% → 97.1%

Failed request / error-handling test results

12 / 16 handled, 0 crashes

Accessibility or usability evaluation result

Not measured

Value and Demo Roadmap

Pilot with planners before live operational deployment

WHO USES IT

CEB energy planners
Grid managers & utility staff
Policy and research teams

VALUE CREATED

Plan with uncertainty ranges
Review abnormal days faster
Compare models transparently
Build evidence for policy decisions

PILOT PATH

1 Historical-data validation
2 Planner usability feedback
3 Controlled near-real-time trial
4 Live integration after governance approval

LIVE DEMO STORY

1. Choose date / inputs
2. Generate forecast
3. Anomaly Detection
4. Explainable AI
5. Decision Support

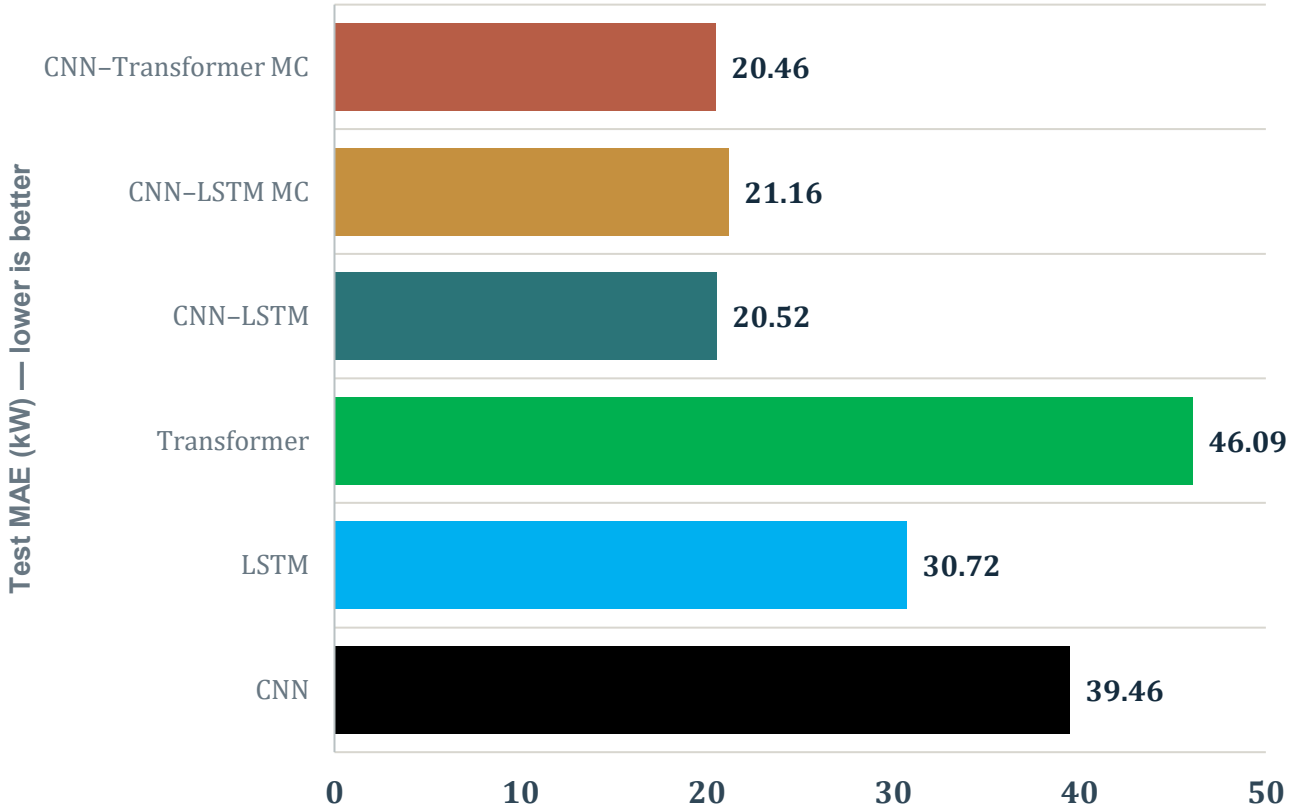
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Adaptive Forecasting and Prediction System

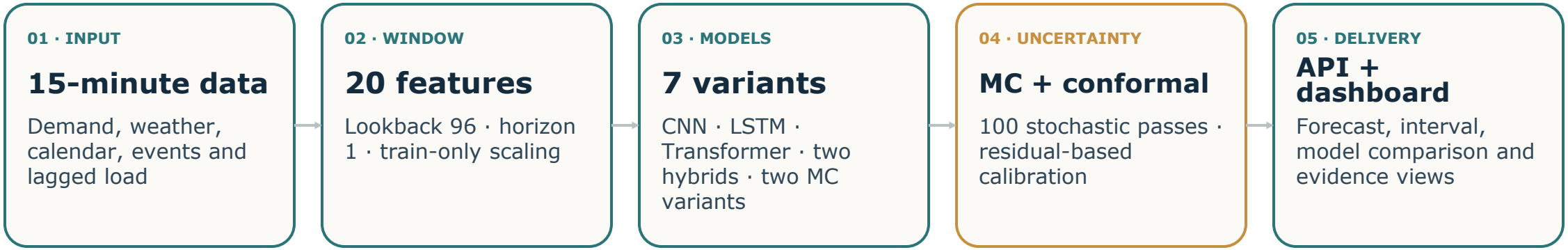
Specialization: Information Technology

Component : Adaptive Forecasting and Prediction System

Comments: Do Model Comparison



Workflow



IMPLEMENTED

Shared preprocessing and scalers
Seven model artefacts

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Behaviour Intelligence

Specialization: Information Technology

Component : Behaviour Intelligence

Comments: *Feedback from Progress Presentation 01 and how it was addressed*

COMMENT RECEIVED	ADDRESSED IN THIS ITERATION
01 *Find the total number of days and analyse the demand; also predict future behaviours with matching patterns.*	**1,977 complete days** analysed from 189,888 fifteen-minute readings (2020-01-01 to 2025-05-30). Future behaviour predicted by month and weekday pattern matching, served live at **/api/predict-future-behavior**.
02 *Moderate idea; K-Means clustering used to identify peak days and abnormalities.*	**Clarified into three separate mechanisms.** K-Means (k=2) clusters profile shape only. Peak days come from percentile rules (**212 days**); abnormal days from Isolation Forest (**60 flagged, 21 labelled**).
03 *Do a model comparison on 2–3 models on model performance.*	**Eight models compared** — four clustering (K-Means, DBSCAN, Agglomerative, Gaussian Mixture) and four anomaly detectors (Isolation Forest, One-Class SVM, Local Outlier Factor, Z-score). Both ranking criteria were re-derived after the original ones were found circular.

Status Pipeline, models, API and dashboard complete · 10 endpoints live · clustering and anomaly comparison pages rebuilt with corrected ranking methodology

Workflow

01 · INGEST

189,888 rows

15-minute readings · Jan 2020 to May 2025

02 · PROFILE

96 slots/day

Pivot to daily profiles · 1,977 complete days

03 · MODELS

3 models fitted

K-Means · DBSCAN · Isolation Forest on 96-pt profiles

04 · LABEL

4 day labels

Percentile peak rules + anomaly flags, by priority

05 · SERVE

10 API endpoints

FastAPI service + three-tab React dashboard

WHAT EACH STAGE PRODUCES

cleaned_data.csv · 189,793 cleaned rows at 15-minute resolution

daily_profiles.csv · 1,977 days × 96 load slots

kmeans / dbscan / isolation_forest .pkl · plus the fitted profile scaler

behavior_labels.csv · Normal, Peak, Abnormal, Holiday / Low

IMPLEMENTED

Full pipeline runs end to end from raw CSV to dashboard

Clustering and anomaly model comparison pages served live

Future behaviour prediction by pattern matching

Results and Model Comparison

DAY CLASSIFICATION — all 1,977 days, 2020-01-01 to 2025-05-30

1,279 64.69%	**465** 23.52%	**212** 10.72%	**21** 1.06%
Normal weekday	Holiday / low	Peak demand	Abnormal

CLUSTERING — ranked on mean of three metric ranks

Model	Silhouette	DBI	CH	Mean rank
K-Means (selected)	**0.4729**	**0.8373**	**1367.33**	**1.667**
Agglomerative	0.4536	0.7929	1324.10	2.000
Gaussian Mixture	0.6558	2.9566	58.46	2.333
DBSCAN	—	—	—	n/a

ANOMALY DETECTION — ranked on bootstrap stability

Model	Flagged	Rate	Mean Jaccard
Isolation Forest (selected)	**60**	**3.03%**	**0.7769**
Local Outlier Factor	60	3.03%	0.7769
One-Class SVM	75	3.79%	0.2812
Z-score	0	0.00%	n/a

WHY THE RANKING CRITERIA CHANGED

Clustering. A Silhouette-only ranking selects Gaussian Mixture, which is worst on both other metrics. Mean of ranks selects K-Means — the model the pipeline actually uses.

Anomaly detection. Three of the four detectors are pre-tuned to a 3% rate, so ranking on closeness to 3% rewards the hyperparameter, not the detector.

FINDINGS WORTH REPORTING

Z-score removed. Max |z| is 2.50 and 2.93 against a threshold of 3.0 — it can never fire on this data.

DBSCAN reported as a negative result. No eps gives both two clusters and low noise; the density structure is unimodal.

Isolation Forest vs Z-score: ρ = 0.9028. Profile-shape detection tracks aggregate magnitude closely on this dataset.

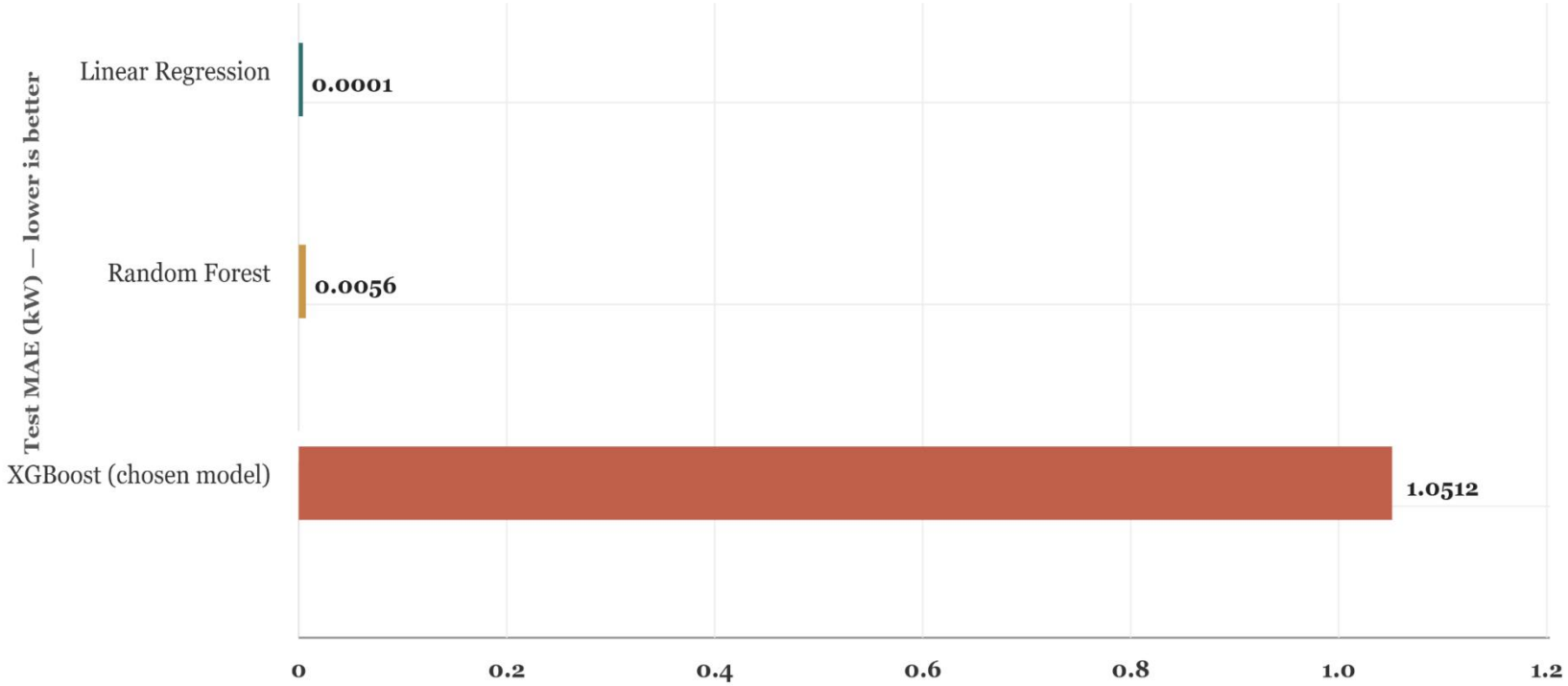
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XAI: LIME & SHAP Analysis

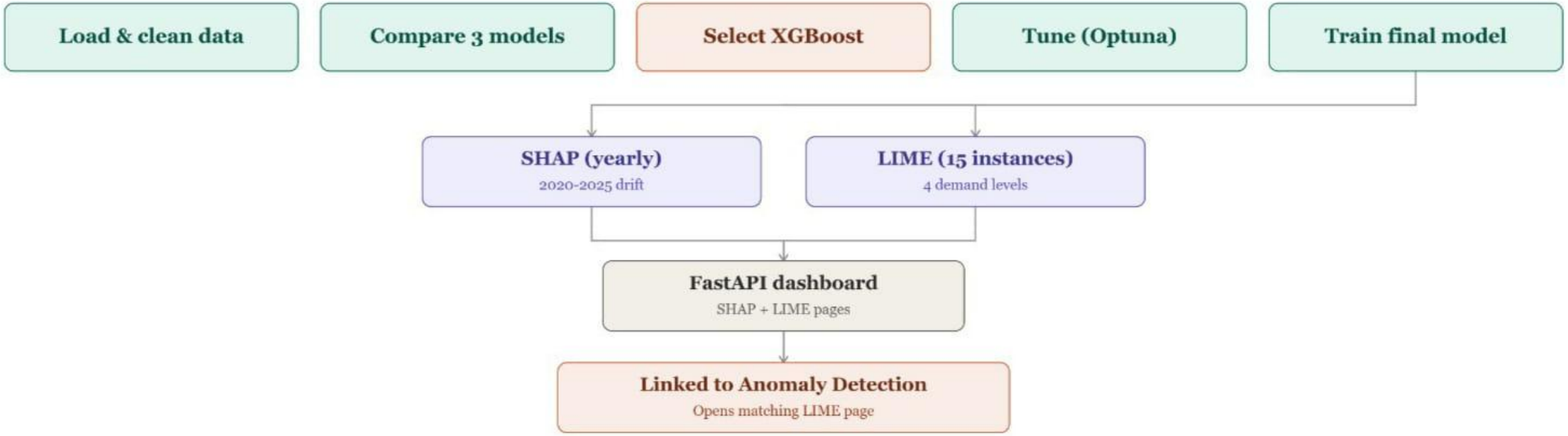
Specialization: Information Technology

Component : Explainable AI for Electricity Demand Forecasting

Comments: Do a model comparison with 2-3 models



Workflow


Load & clean data
Compare 3 models
Select XGBoost
Tune (Optuna)
Train final model
SHAP (yearly)
2020-2025 drift
LIME (15 instances)
4 demand levels
FastAPI dashboard
SHAP + LIME pages
Linked to Anomaly Detection
Opens matching LIME page

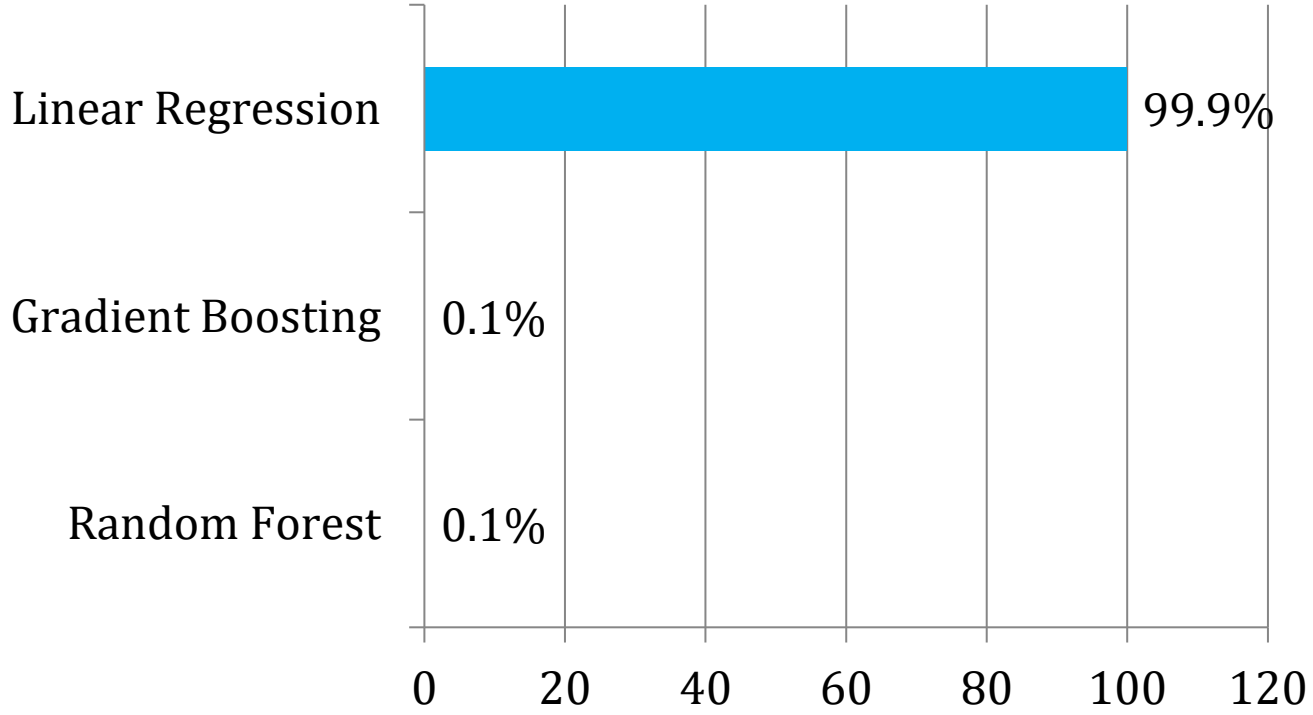
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Scenario Analysis & Decision Support System

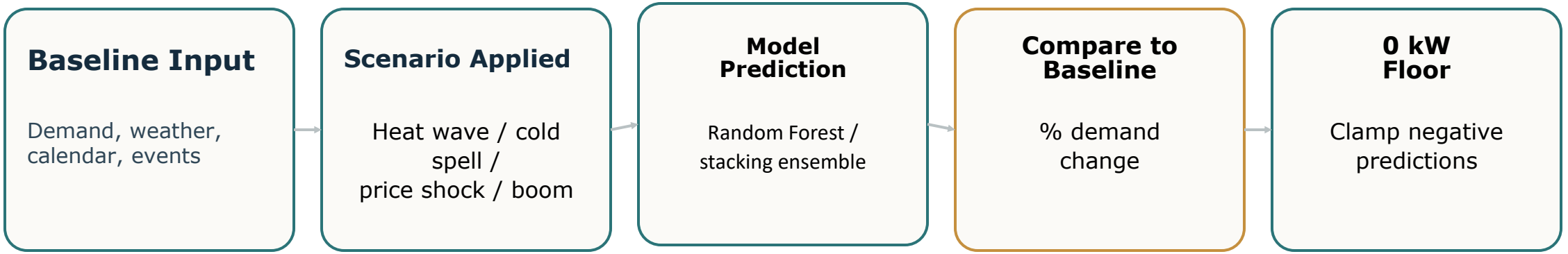
Specialization: Information Technology

Component : Scenario Analysis & Decision Support System

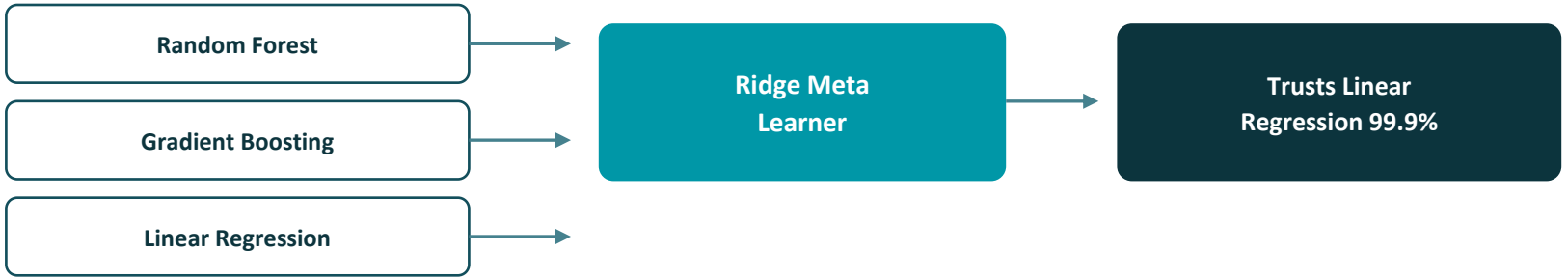
Comments: Model comparison completed + deployment risk tested and fixed



Workflow



Behind "Model Prediction": a tuned stacking ensemble



No data leakage — trained on 5-fold out-of-fold predictions, never the base learners' own training rows.

Thank You